

Deep Patel

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SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C++
Frameworks: Spring Boot, Spring Security, Next.js, React, Node.js, Express, Flask, FastAPI
Databases: PostgreSQL, MongoDB, MySQL, Redis, Vector DBs (pgvector, Chroma)
Tools & DevOps: Git, GitHub Actions, Docker, Kubernetes, REST APIs, JWT, Prisma, Clerk, PyTest, Jest
Cloud & Infrastructure: AWS (EC2, S3, Lambda), Google Cloud, Terraform, Linux, OpenShift
Big Data: Apache Spark, PySpark, Spark SQL, MLlib, Hadoop, BigQuery, Apache Kafka
AI & LLMs: OpenAI, Google Gemini, RAG, Prompt Engineering, LangChain, Vector Search

EXPERIENCE

BU Spark! (Boston University) Sep 2026 – Present

Technical Product Manager

- Lead delivery for 3 client software projects, turning each client brief into milestones, sprint plans, and weekly technical reviews for the student engineering teams.
- Scoped the rebuild of a Boston heating-system map covering nearly 100K parcels into a Vite, TypeScript, and MapLibre app that loads only the parcels in the current view.

BU Spark! (Boston University) May 2026 – Jul 2026

Software Engineer Intern

- Replaced phone and spreadsheet-based orientation registration for 7 regional child care agencies by building a web platform for the Massachusetts Department of Early Education and Care.
- Built the app in Next.js, TypeScript, PostgreSQL, Prisma, and Clerk, with provider registration, attendance tracking, and separate views for providers, regional staff, and state administrators.
- Wrote agency-scoped access controls so each CCR&R agency sees only its own records, while EEC administrators keep statewide reporting.

Prama (AI, Cloud & Data Engineering Consultancy) Jan 2025 – Aug 2025

Software Engineer

- Shipped a React, Spring Boot, and MongoDB dashboard that put every team project in one place, with Microsoft-auth role views: admins see everything, managers staff their own projects, and contributors see only assigned work.
- Added one-click EC2 and Lambda start/stop controls to the dashboard, so managers could run AWS resources without opening the AWS Console.
- Cut MongoDB and PostgreSQL query latency by 35% by adding compound indexes and a Redis cache layer, validated in load tests at 500+ requests/sec.
- Dropped endpoint failures from ~12% to under 0.5% in production by tracing them to token refresh and adding a retry-backoff mechanism.

PROJECTS

Gateway Cities Dashboard | *React, Vite, Flask, Pandas, Leaflet, Census API, Google Gemini, RAG* 2026

- Served demographic data for 260 Massachusetts places in under 100ms per response through a Flask REST API with 13 parameterized endpoints backed by LRU-cached Parquet reads.
- Built a React/Vite frontend with Leaflet choropleth maps over Census TIGERweb GeoJSON, giving users city-level filters and country-of-origin drill-downs across 26 Gateway Cities.

Justice Lens: Boston Criminal Justice Analytics | *Python, Streamlit, Pandas, Plotly* 2026

- Kept page transitions under 1 second in a multi-page Streamlit app over 100K+ Boston PD records (2011 to 2020) by caching data transforms and rendering charts with Plotly.
- Linked officer records across datasets with no shared key using fuzzy name normalization, and merged 40+ variant allegation labels to surface repeat-complaint patterns across 10 years of data.

DataSentinel: Agentic Data Quality Platform | *FastAPI, React, Kafka, GPT-4o-mini, Pydantic, MLflow* 2026

- Built a multi-agent LLM system (GPT-4o-mini, Pydantic-validated JSON) where a coordinator routes each data failure to specialized agents that classify it into 5 categories, repair bad records, and open GitHub issues on schema violations.
- Shipped it behind a FastAPI backend and React dashboard showing live failure rates, agent accuracy, and cost per run, with a PostgreSQL prompt registry that rolls back automatically if accuracy drops below 85%.

EDUCATION

Boston University Expected Dec 2026

M.S. Data Science, Data Engineering at Scale, ML for Business Analytics

CUNY Queens College Dec 2024

B.A. Computer Science, Data Structures, Databases, Operating Systems, Software Engineering